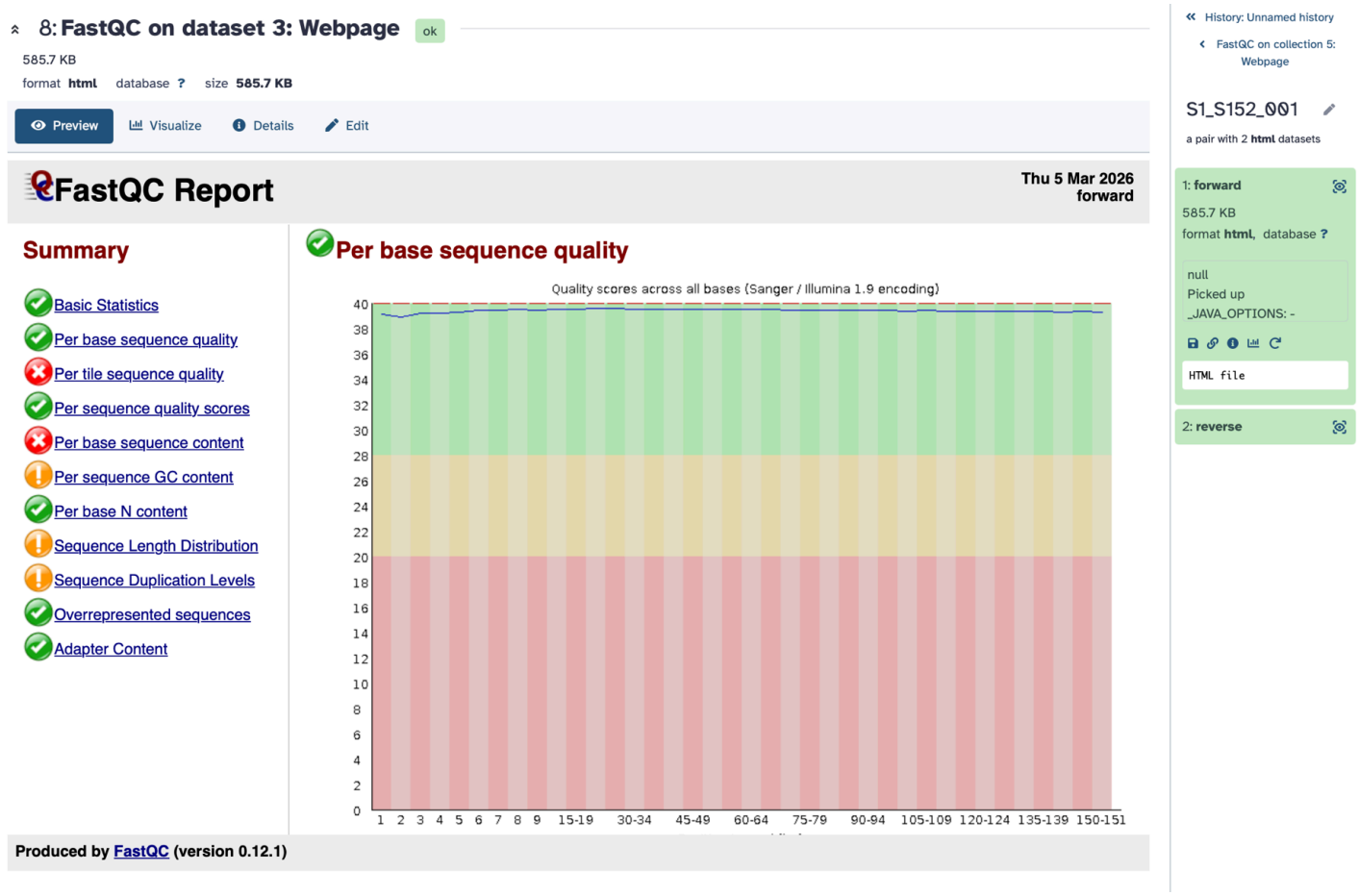
Upload screenshot: FASTQC quality plot (raw)

Choose File QualityPl…RawFastQC



Step 2 — Trim and filter (fastp)

In Galaxy: 1. Run **fastp** on your raw reads (default settings are fine unless instructor says otherwise).
2. Save the **trimmed FASTQ** output and the fastp report (HTML/JSON) for your records.

Questions (fastp)

2A. What did fastp remove/trim? (adapters, low-quality tails, short reads, etc.)

FASTP removed reads with low quality, too many N, those which were too short, and reads with adapter dimers.

2B. Did read length / counts change after trimming? If so, how?

Mean read length did not change after filtering, but some bp appear to have been trimmed as the initial count was 463.553676 M then 458.080025 M after trimming.

(You do not need to upload the fastp HTML; keep it for your workflow folder.)

Step 3 — FASTQC again (after trimming)

In Galaxy: 1. Run **FASTQC** on the trimmed reads.
2. Inspect the report.

Questions (FASTQC — trimmed)

3A. Did overall quality improve compared to raw?

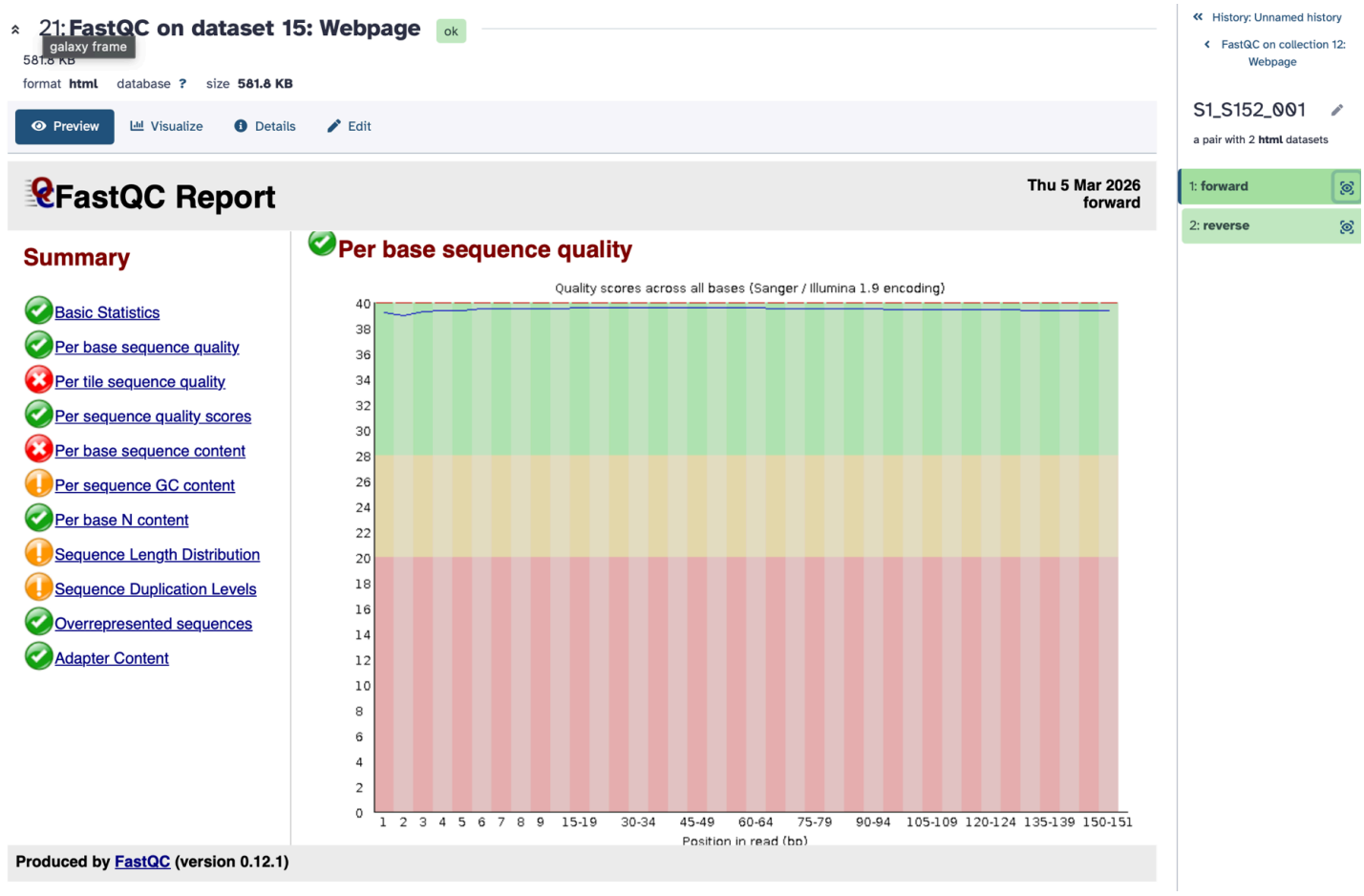
Quality is actually about the same!

3B. Are adapters still present?

There still appear to be little to no adapters present after trimming.

Upload screenshot: FASTQC quality plot (trimmed)

Choose File QualityPlo…medfastqc



QUAST version:

QUAST 5.3.0

C) Write your workflow notes (copy/paste into `workflow_day12.md`)

Use your sampleID and versions.

```
# Day 12 — Galaxy QC → Trim → Assembly → QUAST

## Sample
- sampleID: YOUR_SAMPLEID
- raw data: (where it came from / dataset name)

## Tools (Galaxy / usegalaxy.org)
- FastQC: VERSION
- fastp: VERSION
- SPAdes: VERSION
- QUAST: VERSION

## Steps
1. FastQC on raw reads → saved report as: YOUR_SAMPLEID_raw_fastqc.html
2. fastp trimming (default settings) → saved report as:
YOUR_SAMPLEID_fastp_report.html
3. FastQC on trimmed reads → saved report as: YOUR_SAMPLEID_trimmed_fastqc.html
4. SPAdes assembly on trimmed reads → saved contigs as:
YOUR_SAMPLEID_spades_contigs.fasta
5. QUAST evaluation on contigs → saved report as: YOUR_SAMPLEID_quast_report.pdf

## Notes / interpretation
- (1–3 bullets on what changed after trimming)
- (1–3 bullets on what QUAST suggests about contiguity)
```

Export (PDF) & Submission

1. Click **Prepare submission preview** (generates a clean summary).
2. Click **Download as PDF (Print)** and save.
3. Commit your downloaded/renamed files + `workflow_day12.md` to your Git repo.

Submission preview

Setup

Galaxy account: ariellenames
Raw datasets: Sample1_R1.fastq.gz, Sample1_R2.fastq.gz
Local folder: day12_galaxy_AssemblyandAssessment
sampleID: Unknown1

FASTQC (raw):

```
Read length: 35-151 bp
Quality pattern: Quality does appear to drop toward the end of reads a little bit but
does so more near the beginning.
Adapters: It would appear that they are not, as the graph shows one line for PolyG at
zero on the Y-axis across corresponding X-axis values (read position).
```

fastp:

```
Removed/trimmed: FASTP removed reads with low quality, too many N, those which were
too short, and reads with adapter dimers.
Read counts change: Mean read length did not change after filtering, but some bp
appear to have been trimmed as the initial count was 463.553676 M then 458.080025 M
after trimming.
```

FASTQC (trimmed):

```
Quality improvement: Quality is actually about the same!
Adapters after trimming: There still appear to be little to no adapters present after
trimming.
```

SPAdes:

```
Downloaded contigs file: Unknown1_contigs.fasta
```

QUAST:

```
N50: 193475
L50: 8
# contigs: 79
Total length: 4797060
```

Interpretation:

```
Trimming effect: Although it seems that little trimming occurred, it is likely to
have improved assembly quality, as the N50 value was good-- about 193 kb.

Fragmentation: Coverage is decently high, so I would assume that the assembly isn't
too fragmented.
```

Versions (from Galaxy Job Info):

```
FastQC: FastQC version 0.12.1
fastp: fastp 1.1.0
SPAdes: SPAdes 4.2.0
QUAST: QUAST 5.3.0
```

Reproducibility checklist:

```
FASTQC raw downloaded+renamed: Yes
fastp report downloaded+renamed: Yes
FASTQC trimmed downloaded+renamed: Yes
SPAdes contigs downloaded+renamed: Yes
QUAST PDF downloaded+renamed: Yes
workflow_day12.md created: Yes
```